

Where the Number Helix Lives, and How to Find It: Cross-Architecture and Cross-Script Geometry of Integer Representations

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Abstract

Kantamneni and Tegmark [1] showed that for an integer $a \in [0, 99]$, the residual stream of three mid-sized LLMs (GPT-J-6B, Pythia-6.9B, Llama-3.1-8B) encodes a as a generalised helix—a linear “number-line” axis plus four orthogonal modular circles at periods $T \in \{2, 5, 10, 100\}$, and that this helix is causally used to perform addition. Their basis itself is the LLM-scale analogue of Nanda et al. [2], who reverse-engineered the same trig solution in a small transformer trained from scratch on modular addition—the canonical grokking task of Power et al. [3]. We extend the LLM result along three orthogonal axes. **Architecture:** we add a new model family (Gemma 4, at E4B and 31B) and re-measure Pythia and Llama under a wider protocol, recovering the helix on all four models but at radically different depths—Pythia/Llama mid-stack, Gemma-4-E4B at $L=5$ of 30, Gemma-4-31B as a sharp spike at $L=29$ of 61. **Script:** across eight numeral systems we find a three-way split—positional base-10 scripts the model has been trained on (Latin, Devanagari, Arabic-Indic, Persian, positional Chinese) reproduce the helix on multilingual models; additive systems (Roman, Greek alphabetic) collapse it in two qualitatively distinct ways distinguishable only by inspecting where PC1 and helix R^2 peak in layer-space; and Babylonian cuneiform, a positional base-60 system, encodes its native period 60 on *all four models*, provided we widen the input window past one base-60 wrap and add $T=60$ to the trig basis (adding the single missing period recovers 7–13 percentage points of variance uniformly across the four models). 2-D PCA scatters of the residual stream further reveal that on Gemma the first principal component is not magnitude but *tokenisation* (PC1 captures up to 96.6% of variance but separates points by digit count, not value). We extract two methodological lessons: (M1) any basis-dependent metric must be cross-checked against a basis-free FFT and against a 2-D PCA scatter; (M2) the input range must contain several wraps of every candidate period.

1 Introduction

Mechanistic interpretability has produced a growing catalogue of “cognitive” structures embedded in transformer residual streams—induction heads, indirect-object identification circuits, and so on. Arithmetic has been a particularly load-bearing case: Power et al. [3] observed that a small transformer trained on modular addition continues training long past memorisation before suddenly generalising (“grokking”), and Nanda et al. [2] reverse-engineered the resulting solution as a Fourier/trig basis—integers laid out on circles, addition implemented via $\cos(a+b) = \cos a \cos b - \sin a \sin b$. The number-helix of Kantamneni and Tegmark [1] carries the same basis into *pretrained* LLMs on the naturals: integers are not stored as opaque embeddings but laid out as a 3-D spiral—a linear number-line axis plus four orthogonal circles—and addition is performed by rotating the spiral’s circular components. The claim is striking; it is

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also based on three models in the same broad decoder-only lineage (GPT-J-6B, Pythia-6.9B, Llama-3.1-8B) and on a single numeral system (Latin digits).

This paper asks: *how general is the helix?* We ask the question along two independent axes:

1. **Architecture:** does the helix extend beyond the decoder-only lineage Kantamneni and Tegmark studied? We add Gemma 4 (E4B and 31B—different tokeniser, different training mix, different depth) and re-measure Pythia and Llama under a wider protocol. If the helix is there, at what depth?
2. **Script:** when the model reads “𐤚”, “𐤛”, “𐤬𐤬𐤬”, does it represent these as numbers in the same helical way, or as ordinary multi-byte words?

Both axes turn out to matter, and both surface methodological subtleties that are easy to miss if one runs only the paper’s exact measurement recipe. Section 2 reviews the helix construction and our two basis-free diagnostics. Section 3 reports the layer-depth findings across four models. Section 4 compares eight numeral systems, and analyses the two qualitatively distinct ways the helix collapses on non-positional scripts. Section 5 contains the main scientific surprise: Babylonian cuneiform is represented at its native base 60 on every model we tested, once the measurement window is widened past a single wrap and the basis is extended with $T = 60$. Section 6 distills the two generalisable methodological lessons.

2 Methods

2.1 Setup

For each integer $a \in [0, n_{\max})$ we feed the string “ $\sqcup\langle\text{numeral}(a)\rangle$ ” to the model and pool the residual stream at a chosen layer L across the sub-tokens of the numeral. With d the hidden width, the result is an activation matrix $H \in \mathbb{R}^{n_{\max} \times d}$. Latin digits 0–99 are single tokens on Pythia, so pooling is trivial there; in every other case (non-Latin scripts; Latin on Llama and Gemma; the $n_{\max}=600$ sweep) we average activations across the numeral’s sub-tokens to avoid a last-byte cycling artifact in which $\{23, 33, 43, \dots\}$ read identical activations because they end in the same glyph.

2.2 The helix basis

The Kantamneni–Tegmark hypothesis is that $h(a)$, for fixed L , factors as

$$h(a) \approx W^\top B(a), \quad B(a) = \left[a, \cos \frac{2\pi a}{T_1}, \sin \frac{2\pi a}{T_1}, \dots, \cos \frac{2\pi a}{T_K}, \sin \frac{2\pi a}{T_K} \right], \quad (1)$$

with $T = \{2, 5, 10, 100\}$ for Latin in $[0, 99]$, giving a $(1+2K) = 9$ -dimensional feature space. The linear coordinate is the “rise” of the helix; each $(\cos, \sin)$ pair places a on the unit circle at angle $2\pi a/T$, so integers sharing the same $a \bmod T$ land on the same point. Addition becomes rotation of angles: the paper’s “Clock” algorithm.

We solve $H \approx BW$ by ordinary least squares and report the variance-weighted coefficient of determination, which we call *helix* R^2 .

2.3 Three diagnostics

Helix R^2 alone is fragile—it depends on the fixed basis T . We pair it with two basis-free diagnostics.

Fourier diagnostic. For each hidden dimension j , the column $H_{:,j}$ is a sequence indexed by a . We take its discrete Fourier transform, average the magnitudes across j , and inspect the spectrum. Periodic structure *at any period*, including ones not in T , surfaces as a peak. We label the top peaks via `scipy.signal.find_peaks` so that the FFT panel honestly reports whatever periods the model actually encodes, rather than only those the paper’s basis assumes.

PC1 diagnostic. The leading principal component of H is the direction of maximum variance. If the helix’s linear axis is present, plotting PC1 against a should yield a clean line; we report the linear-fit R^2 as *PC1 R^2* .

Eckart–Young upper bound. Helix R^2 uses a 9-D fit, but 9 is also the dimensionality of many other regression targets. The best possible 9-D fit is the 9-D PCA reconstruction; we call its variance-weighted R^2 the *K-D PCA bound*, and report the ratio $\rho = R^2_{\text{helix}}/R^2_{\text{PCA}}$ as *helix/PCA*. When $\rho \gtrsim 0.8$, the trig basis is *the* 9-D structure of the residual stream rather than merely *a* 9-D approximation of it.

2.4 Layer sweep

The paper reads at layer $L = N/2$ in an N -layer model. We capture activations at all $L = 0, 1, \dots, N$ in one forward pass per integer and compute (PC1 R^2 , helix R^2 , K-D PCA bound) at each. The peak helix- R^2 layer is then used for the standard regression fit. As Section 3 shows, “read the middle layer” is a Pythia/GPT-J convention, not a universal one.

2.5 2-D PCA scatter

At the helix-peak layer of each cell we scatter the first two principal components of H , colour-coded by a , with consecutive integers connected by a faint trajectory line. This is the cheapest basis-free diagnostic for *non*-periodic structure (Roman’s threshold-aligned clusters; Greek’s diffuse noise; Gemma’s tokenisation-dominated PC1; Section 4.5).

3 Architecture: the helix is consistent, the layer is not

We test four base (non-instruction-tuned) models: Pythia-6.9B, Llama-3.1-8B, Gemma-4-E4B (efficient ~ 4 B), and Gemma-4-31B. All four converge on the same geometric encoding on Latin digits 0–99 (Table 1), but the depth at which the helix peaks varies dramatically, and so does the shape of the depth profile (Figure 1).

Table 1: Latin helix metrics at the peak layer of each model. “helix/PCA” is the ratio of helix R^2 to the 9-D PCA upper bound (values $\gtrsim 0.8$ indicate the trig basis dominates the 9-D subspace).

Model	helix R^2	helix/PCA	peak layer	total layers
Pythia-6.9B	0.53	0.85	15	32
Llama-3.1-8B	0.63	0.85	14	32
Gemma-4-E4B	0.50	0.77	5	~ 30
Gemma-4-31B	0.70	0.75	29	61

Three depth regimes emerge:

- **Pythia and Llama:** helix forms in the back half of the stack and remains until at least $L = N - 1$. Either $L = N/2$ or $L = N - 1$ would have worked.



Figure 1: $PC1 R^2$, helix R^2 and helix/PCA ratio plotted against layer index, for each of the four models (Latin digits 0–99). Pythia and Llama rise into the back half of the stack and stay high. Gemma-4-E4B peaks early and decays. Gemma-4-31B is nearly flat at ~ 0.5 with a sharp spike at $L=29$ that reads as “no helix” if one samples at $L=30 = N/2$.

- **Gemma-4-E4B:** helix is strongest at $L = 0-5$ and decays in later layers. The paper’s middle-layer recipe would literally read at decayed structure.

- **Gemma-4-31B:** helix R^2 is essentially flat at ~ 0.5 across the whole stack, with a single sharp peak of 0.70 at $L = 29$. The default $L = N/2 = 30$ misses the peak by exactly one layer.

We initially concluded “Gemma-4-31B does not form a helix” from a default-layer reading—a conclusion wrong by a single layer. The methodological consequence is unambiguous: *for any new model, a layer sweep is mandatory*. The helix is general; the recipe for finding it is not.

4 Script: positional structure is necessary, training exposure is what makes the helix appear

4.1 The numeral systems at a glance

For readers unfamiliar with the non-Latin systems we test:

- **Arabic-Indic** (٠١٢٣٤٥٦٧٨٩) and **Persian** (٠١٢٣٤٥٦٧٨٩) are positional base-10 like Latin, differing only in glyph shape (Persian is a visual variant of Arabic-Indic). Numerals are written most-significant-first despite Arabic script flowing right-to-left.
- **Devanagari** (०१२३४५६७८९) is also positional base-10, used across Indian languages including Hindi, Sanskrit, and Marathi.
- **Positional Chinese** uses the digits 零一三三四五六七八九 in the same place-value layout as Latin (e.g. 二三 for 23). Classical Chinese also admits a non-positional form (二十三, “two-tens-three”); we do not use it here.
- **Greek alphabetic (Milesian)** is additive: each letter carries a fixed value ($\alpha = 1, \dots, \theta = 9, \iota = 10, \kappa = 20, \dots, \pi = 80, \rho = 90, \rho = 100, \dots$). 23 is $\kappa\gamma = 20 + 3$. There is no shared “ones digit” between 13, 23, and 33.
- **Roman** is additive with subtractive shortcuts at 4, 9, 40, 90, etc.: I= 1, V= 5, X= 10, L= 50, C= 100. 23 = XXIII. Glyphs shift at $4 \leftrightarrow 5$, $9 \leftrightarrow 10$, $49 \leftrightarrow 50$, $89 \leftrightarrow 90$ — the same boundaries that later appear as staircase discontinuities in the PC1 panel (Section 4.4).
- **Babylonian cuneiform** is positional at base 60, additive within each sexagesimal column. Two glyphs combine inside a column: 𐎶 (one) and 𐎵 (ten). 23 is 𐎶𐎵𐎶 — two tens plus three ones, all in one column. 60 opens a new column to the left (Section 5).

4.2 Setup

We rendered every integer in $[0, 99]$ in eight numeral systems (Table 2) and re-ran the analysis under mean-pooling.

4.3 Helix/PCA across scripts

Three observations stand out:

Non-positional scripts collapse uniformly. On every model, Greek and Roman collapse the helix/PCA ratio to ~ 0.55 – 0.65 . A trig basis with a tens-loop only buys representational efficiency if the numeral system has a tens column. Roman IV versus V is not the kind of transition a fixed-period circle can align to.

Latin is uniform. All four models clear 0.75 on Latin. The shape itself is not model-specific.

Table 2: Numeral systems tested. The Babylonian column is the most interesting and is treated separately in Section 5.

Script	Example (23)	Family
Latin	23	positional base-10
Arabic-Indic	٢٣	positional base-10
Persian	۲۳	positional base-10
Devanagari	२३	positional base-10
Chinese (pos.)	二三	positional base-10
Greek alphabetic	κγ	additive (Milesian)
Roman	XXIII	additive
Babylonian	◀◀III	positional base-60, additive within column

Table 3: Helix/PCA ratio at each model’s peak layer, by script (mean pooling). Bold = highest score per model.

Script	Pythia-6.9B	Llama-3.1-8B	Gemma-4-E4B	Gemma-4-31B
Latin	0.85	0.85	0.77	0.75
Arabic-Indic	0.55	0.58	0.66	0.69
Persian	0.55	0.70	0.72	0.72
Devanagari	0.53	0.61	0.78	0.79
Chinese	0.60	0.58	0.68	0.66
Greek (additive)	0.54	0.57	0.57	0.57
Roman (additive)	0.54	0.55	0.62	0.64

Non-Latin positional cells split by training data. Pythia on Devanagari (0.53) is statistically indistinguishable from Pythia on Greek (0.54): the helix is gone. Llama on Devanagari (0.61) is a little better. Gemma-4-31B on Devanagari hits 0.79, matching its own Latin score of 0.75. Pythia’s training mix is English-heavy; Llama saw more multilingual data; Gemma is the most multilingual of the four. *Positional structure is necessary for the helix, but training exposure to the specific script is what makes it appear.* The encoding is built per-script during pre-training; architecture alone is not enough.

4.4 Two qualitatively distinct collapses on non-positional scripts

The naive reading of Table 3 is that “Greek and Roman collapse the helix.” That is correct on the numerical metric but misleading: the underlying geometry is qualitatively different in the two cases. Reading PC1 R^2 alongside helix R^2 , *and the layer at which each peaks*, reveals the difference (Table 4).

Roman: a decomposed encoding, not a fused helix. On Latin, PC1 and helix R^2 peak at the same mid-stack layer; magnitude and periodicity co-locate, and the trig basis literally fits a helix (line $\times$ circles). On Roman, PC1 and helix R^2 peak at *different* layers—9 to 20 apart. Magnitude is present (PC1 R^2 climbs to 0.49–0.66, so the model does coarsely order Roman numerals), and so is periodic structure (Figure 2 shows clean FFT peaks at $T = 10$ for the X-cycle and $T = 5$ for the V-cycle), but they live in different parts of the residual stream. Geometrically, Roman is encoded as a *number-line module* at one depth plus a *letter-cycling module* at another, not as a fused helix. The PC1-vs- a panel of Figure 2 is also informative: it is not a straight line but a staircase with discontinuities at the I/V, IX/X, XL/L, and XC/C boundaries—the underlying signal is piecewise linear, not linear, which is why the naive linear-fit $R^2 = 0.23$ understates the magnitude information that is present.

Table 4: Layer of peak PC1 and helix R^2 on Latin, Roman, Greek. Latin: both peak at the same layer (a single helix). Roman: peaks are 9–20 layers apart (decomposed parallel modules). Greek: helix R^2 peaks at $L = 0$ (tokeniser artifact at the embedding).

Model	Script	PC1 R^2 peak	helix R^2 peak	layer gap
Pythia-6.9B	Latin	0.96 @ L5	0.53 @ L32	—
Pythia-6.9B	Roman	0.66 @ L9	0.36 @ L18	9
Pythia-6.9B	Greek	0.10 @ L9	0.28 @ L0	(artifact)
Llama-3.1-8B	Roman	0.51 @ L16	0.36 @ L17	1
Llama-3.1-8B	Greek	0.17 @ L5	0.27 @ L0	(artifact)
Gemma-4-E4B	Roman	0.54 @ L4	0.61 @ L18	14
Gemma-4-31B	Roman	0.49 @ L11	0.64 @ L31	20

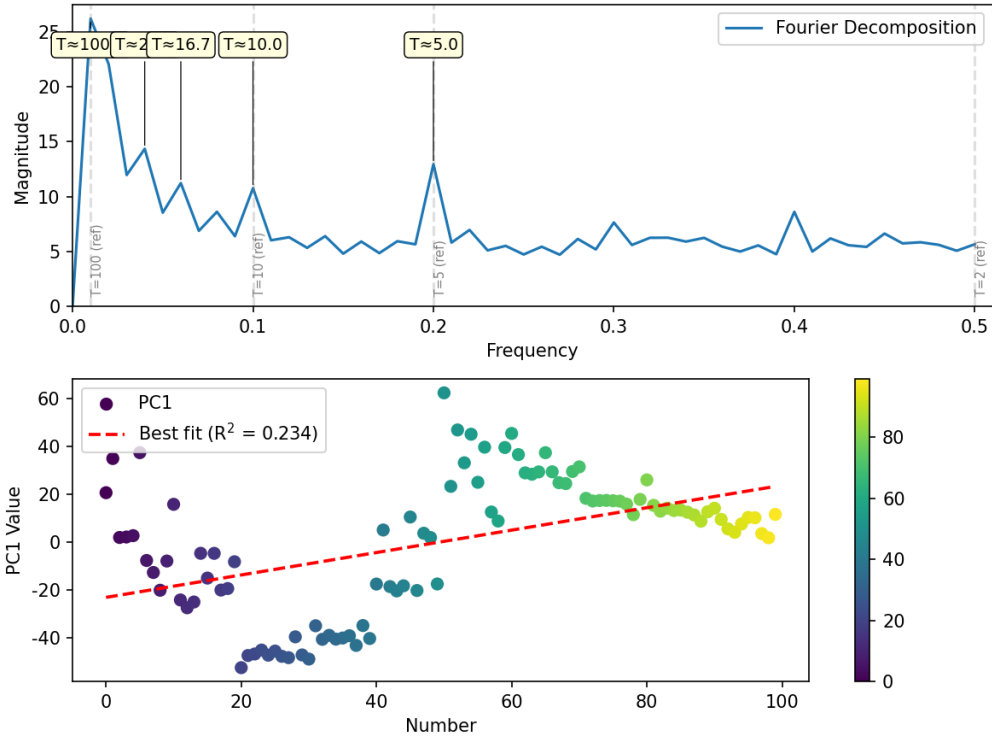


Figure 2: Pythia-6.9B on Roman numerals. **Top:** FFT spectrum shows clean peaks at $T = 10$ (X cycle) and $T = 5$ (V cycle)—the visual periodicity of the additive system. **Bottom:** PC1 versus a is a piecewise function with jumps at the Roman threshold values, not a line. The two components live in different layers (Table 4).

Greek: no quantitative representation at all. PC1 R^2 peaks at 0.10–0.21 on every model; the linear axis never appears. Helix R^2 peaks at layer 0 on three of four models—i.e. at the output of the token embedding, before any transformer block has executed (Figure 3). The FFT magnitudes are ~ 0.20 (compare ~ 30 for Latin); the apparent $T = 10$ peak is a tokeniser artifact in which the second Greek letter cycles through $\alpha, \beta, \gamma, \dots, \theta$ every ten numbers. There is no number representation at any layer for the “helix collapse” to collapse *from*.

The two non-positional collapses are not the same failure mode. Roman is encoded as a sum of separately-stored parts; Greek is not encoded quantitatively at all. The standard “helix R^2 ” metric averages over both modes and reports them as comparable; reading PC1 R^2 and its layer-of-peak alongside helix R^2 is what separates them.

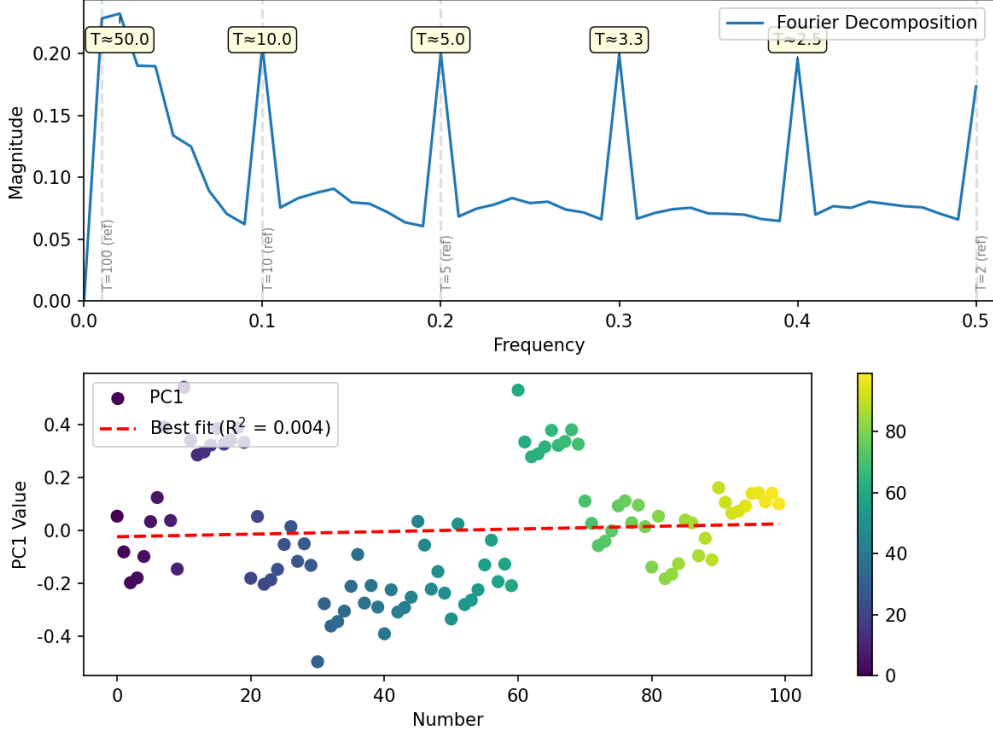


Figure 3: Pythia-6.9B on Greek alphabetic numerals. The FFT magnitude is two orders of magnitude smaller than on Latin; the only periodic structure is a $T = 10$ artifact at $L = 0$ from the alphabetic construction $\kappa + \gamma = \kappa\gamma$. PC1 $R^2 = 0.004$: the model is not representing Greek numerals as quantities.

4.5 2D PCA: cluster structure made visible (and the PC1-is-tokenization surprise)

To probe non-periodic structure that the FFT and the trig-basis fit cannot see, we added a basis-free diagnostic to every cell: a 2D PCA scatter of $h(a)$ at the helix-peak layer, coloured by a , with each numeral labelled and consecutive integers connected by a faint trajectory line. Eight scripts $\times$ four models = 32 panels at `out/<model>/<script>/mean/fig4_pca_2d.png`. Three findings fall out.

Roman: cluster-by-prefix is visible directly. The 2D PCA for Pythia/Roman (Figure 4) shows distinct regions of representation space corresponding to the Roman-numeral prefix classes (I/V, X, L, XC), with the trajectory line zigzagging between clusters at the threshold integers ($4 \rightarrow 5$, $9 \rightarrow 10$, $49 \rightarrow 50$, $89 \rightarrow 90$). Roman’s representation is not a manifold but a partitioning of representation space into prefix classes with crude ordering within each class.

Greek: diffuse noise, confirming the no-representation reading. PC1 + PC2 = 15.5% of total variance on Pythia/Greek (less than half of any other cell) and the points scatter without recognisable shape. There is no learned representation to find; the “helix collapse” was a collapse of nothing.

The Gemma surprise: PC1 is not magnitude; it is token shape. On Gemma-4-31B, helix/PCA = 0.75 and helix $R^2 = 0.70$ on Latin —strong by any measure. But PC1 captures **70.9%** of variance (versus Pythia’s 20.4%) and that variance is along token-shape rather than magnitude (Figure 5). Single-digit numerals (0–9) form one cluster, two-digit numerals starting with “1” (10–19) form another, and 20–99 fill a third dense region. The split is by digit count and leading-digit class. The helix lives in lower-variance directions; PC1 alone would not have found it.

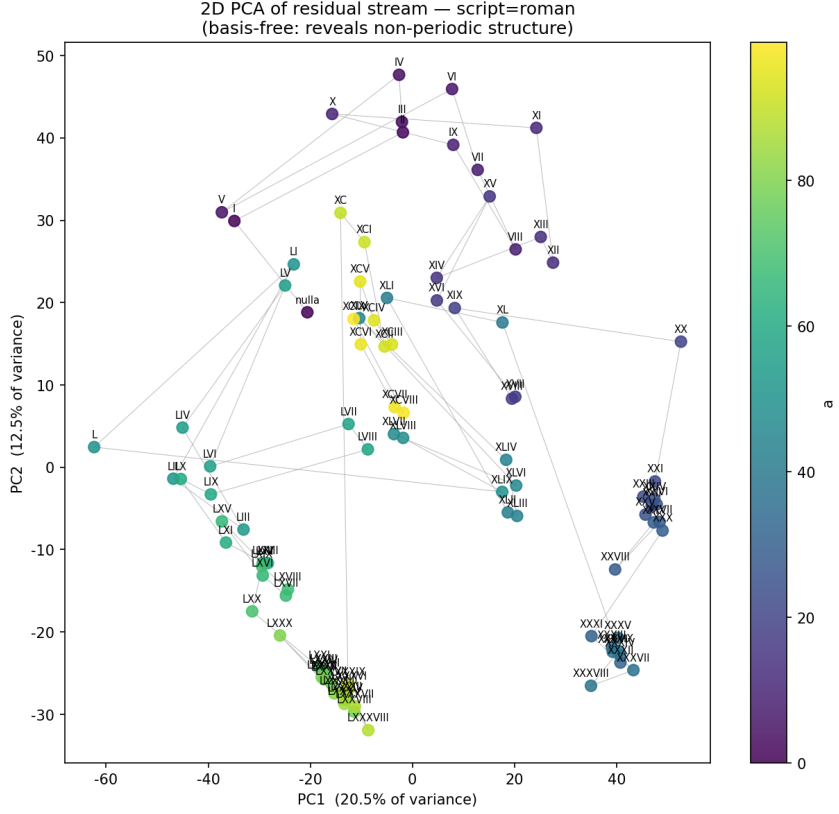


Figure 4: Pythia-6.9B on Roman numerals. 2D PCA at the helix-peak layer reveals discrete clusters by Roman-numeral prefix (I/V, X, L, XC), with the trajectory line zigzagging at threshold integers. The representation is a partition, not a continuous manifold.

The same pattern appears on Devanagari with $PC1 = 94.6\%$ (split between $-$ and $-$), and on Roman at $PC1 = 96.6\%$ (split at the L threshold). In each case $PC1$ captures massive variance, but the variance is overwhelmingly tokenization features, not the number line. In Pythia, by contrast, $PC1$ R^2 and linearity-in- a correspond cleanly: looking at $PC1$ is looking at magnitude.

This carries one specific methodological warning: **a high $PC1$ R^2 is not by itself evidence of a clean magnitude axis**—one must also inspect what $PC1$ separates. In larger or more multilingual models, tokenization features may dominate the first principal component while the helix occupies smaller-variance subspaces. This also partly explains the depth-variance finding of Section 3: Gemma’s helix shares residual-stream space with strong tokenization signals, so it appears as a sharper spike at a specific layer rather than a broad mid-stack plateau.

5 Babylonian: base 60 is there, once you measure properly

Babylonian cuneiform is positional at base 60, additive within each sexagesimal column. The number 23 is <<||| (two tens plus three ones); 60 is !~ (one in the 60s column, a late-period zero placeholder in the ones column); 599 is !||||||| <<<<|||||||. The system has all the ingredients of positional notation, at base 60 instead of base 10.

5.1 The narrow-window measurement says “no”—but cannot

Running the standard $n_{\max} = 100$ protocol with the paper’s basis $T = \{2, 5, 10, 100\}$ gives Table 5.

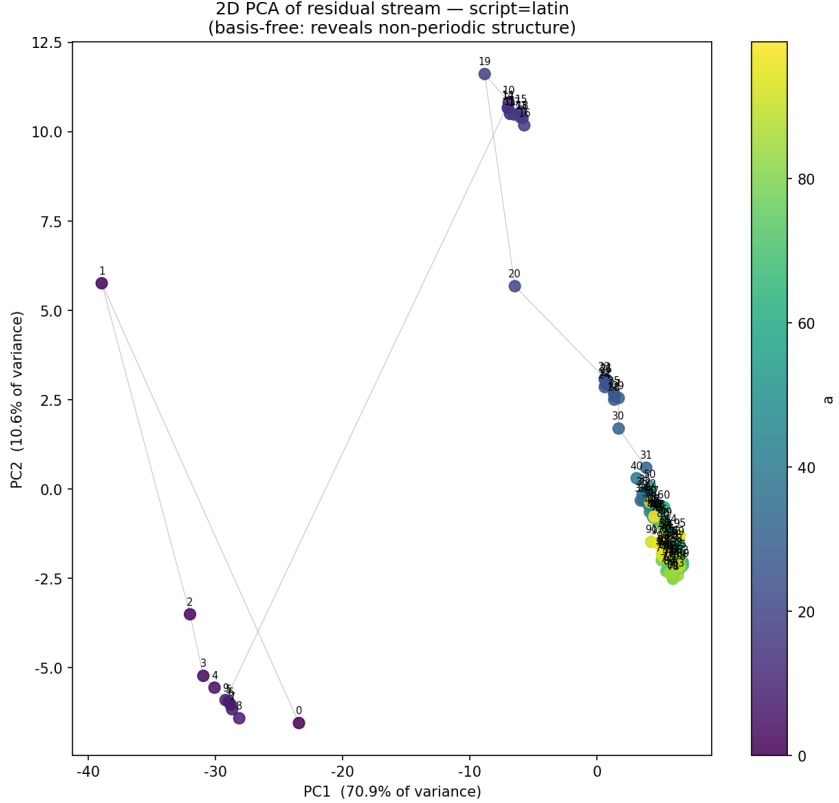


Figure 5: Gemma-4-31B on Latin digits. PC1 captures 70.9% of variance but separates the points by *token shape* (single-digit vs. teens vs. 20–99), not magnitude. Helix $R^2 = 0.70$ persists at this layer, in lower-variance directions PC1 does not reach.

Table 5: Babylonian on the standard 0–99 window, paper basis.

Model	PC1 R^2	helix R^2	helix/PCA
Pythia-6.9B	0.06	0.71	0.74
Llama-3.1-8B	0.10	0.65	0.69
Gemma-4-E4B	0.40	0.75	0.77
Gemma-4-31B	0.51	0.76	0.77

A naive read is that Pythia and Llama have circles but no number-line ($\text{PC1 } R^2 \approx 0$), while Gemma has both. Inspecting the FFT (Figure 6) shows labelled peaks at $T = 10$ and $T = 100$, but *no* $T = 60$. The PC1 panel shows two clusters with a hard step at 60—the wrap-boundary is encoded as a categorical state change, not as a rotation.

This is the measurement artifact: in $[0, 99]$, base 60 wraps exactly *once*, at $a = 60$. A single discontinuity is not a periodic signal an FFT can localise. “No base-60 structure” is, at this $n_{\max}$, indistinguishable from “structure the window cannot resolve.”

5.2 Widening the window: $T \approx 60$ emerges in the FFT

At $n_{\max} = 600$ —ten full sexagesimal wraps—the picture changes sharply (Table 6). Every model now has a real magnitude axis ($\text{PC1 } R^2 \approx 0.55\text{--}0.69$); Pythia’s and Llama’s previous $\text{PC1 } R^2 \approx 0$ was an artifact of one wrap.

The FFT at $n_{\max} = 600$ shows a clean labelled peak at $T \approx 60$ (Figure 7). For Llama and Gemma-4-31B, base-60 structure appears mostly via its second harmonic at $T = 30$ (frequency

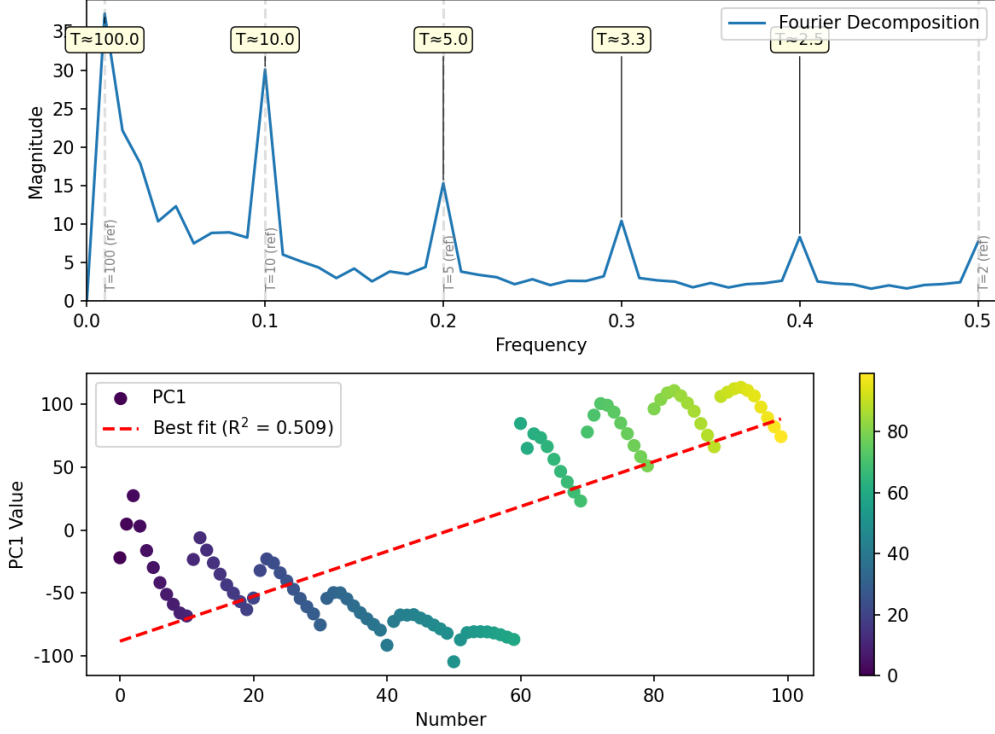


Figure 6: Gemma-4-31B on Babylonian at $n_{\max} = 100$. Top FFT shows $T = 10$ and $T = 100$; *no* $T = 60$. PC1 (bottom) shows numbers 0–59 versus 60–99 as two clusters with a step between them.

Table 6: Babylonian, $n_{\max} = 100 \rightarrow 600$ (paper basis $T = \{2, 5, 10, 100\}$). Both PC1 and FFT diagnostics improve; helix R^2 drops because $T = 60$ is not in the basis.

Model	PC1 R^2 (100 $\rightarrow$ 600)	helix R^2 (100 $\rightarrow$ 600)	helix/PCA
Pythia-6.9B	0.06 $\rightarrow$ 0.55	0.71 $\rightarrow$ 0.66	0.74 $\rightarrow$ 0.69
Llama-3.1-8B	0.10 $\rightarrow$ 0.55	0.65 $\rightarrow$ 0.52	0.69 $\rightarrow$ 0.56
Gemma-4-E4B	0.40 $\rightarrow$ 0.69	0.75 $\rightarrow$ 0.61	0.77 $\rightarrow$ 0.64
Gemma-4-31B	0.51 $\rightarrow$ 0.64	0.76 $\rightarrow$ 0.50	0.77 $\rightarrow$ 0.52

$2/60 = 1/30$, Figure 8). A pure sinusoid at $T = 60$ would have only the fundamental; the presence of strong even harmonics indicates a more sawtooth-like underlying signal, consistent with the per-column reset structure of cuneiform.

So the structure is *present in the FFT*—but helix R^2 on the paper basis *drops* (Table 6), because $T \in \{2, 5, 10, 100\}$ has no column against which to regress period-60 variance. The basis is blind to a frequency the data contains.

5.3 An extended basis recovers 7–13 percentage points of variance, on every model

We re-ran the $n_{\max} = 600$ sweep with $T = \{2, 5, 10, 60, 100\}$ —a single additional period, two additional columns ($\cos \frac{2\pi a}{60}, \sin \frac{2\pi a}{60}$). Table 7 reports the result.

The two models whose FFT showed strongest second-harmonic content (Llama-3.1-8B and Gemma-4-31B) gain the most (+0.107 and +0.133). The regression cannot extract a non-sinusoidal signal from a basis that lacks its fundamental; providing $T = 60$ anchors the phase and the model’s sawtooth-like representation lands on it. PC1 R^2 is unchanged by the basis

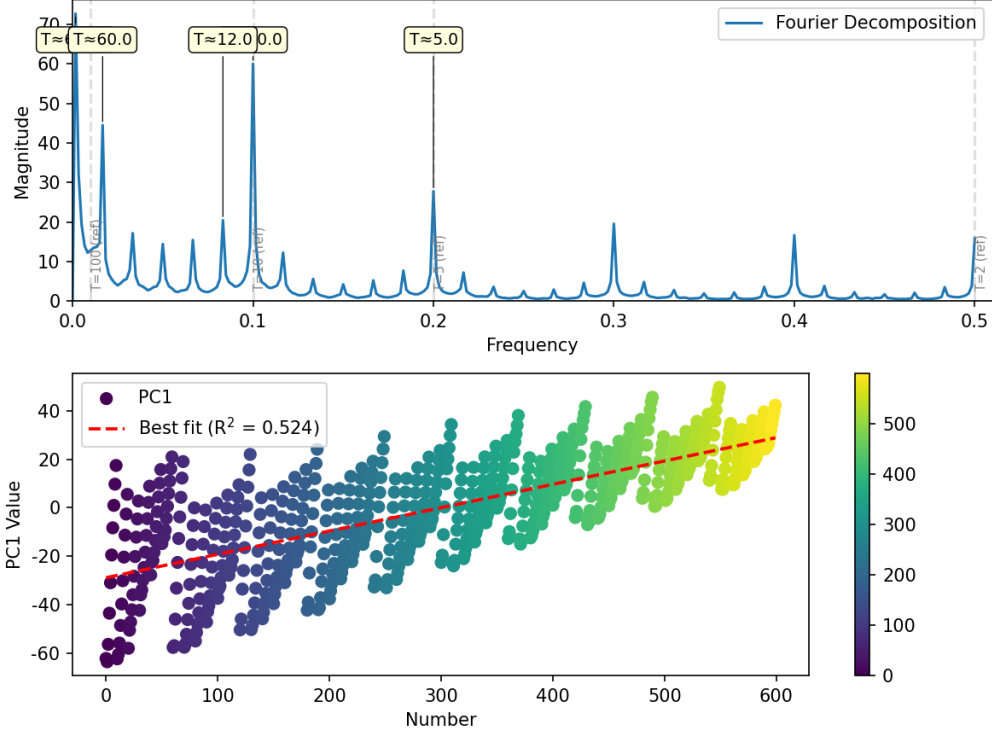


Figure 7: Pythia-6.9B on Babylonian at $n_{\max} = 600$. A clean labelled peak at $T \approx 60$ appears alongside $T = 10$ and various harmonics. The PC1 panel shows the magnitude axis with a per-column sawtooth—the model resets into a new sexagesimal column every 60 integers.

Table 7: Babylonian at $n_{\max} = 600$: paper basis $T = \{2, 5, 10, 100\}$ vs. extended basis $T = \{2, 5, 10, 60, 100\}$. Adding the single missing period recovers 7–13 percentage points of variance on every model.

Model	helix R^2 (paper $\rightarrow$ ext.)	Δ	helix/PCA (paper $\rightarrow$ ext.)	Δ
Pythia-6.9B	0.661 $\rightarrow$ 0.747	+0.086	0.687 $\rightarrow$ 0.772	+0.085
Llama-3.1-8B	0.515 $\rightarrow$ 0.622	+0.107	0.555 $\rightarrow$ 0.645	+0.090
Gemma-4-E4B	0.611 $\rightarrow$ 0.698	+0.087	0.637 $\rightarrow$ 0.699	+0.062
Gemma-4-31B	0.501 $\rightarrow$ 0.634	+0.133	0.523 $\rightarrow$ 0.636	+0.113

change, confirming that no other variable has shifted.

The conclusion is unambiguous:

Base-60 structure is present in the Babylonian representation on all four models, increasing in strength roughly as Pythia < Gemma-4-E4B < Llama < Gemma-4-31B (by Δ helix R^2).

Both measurement adjustments (wider window, extended basis) point the same direction, which is the strongest evidence we have that the structure is real rather than an artifact of either change.

6 Method matters: two general lessons

M1. The basis has bandwidth. Helix R^2 is a fitted regression onto a fixed set of periods. It is sensitive only to structure at those periods; if a model encodes something at a period the

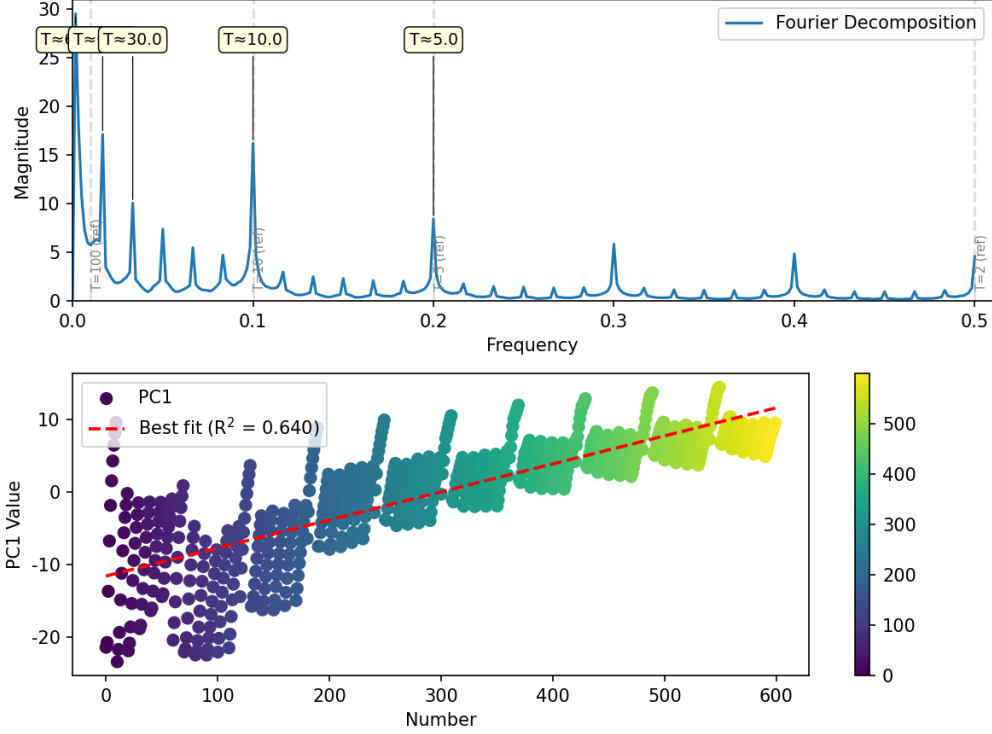


Figure 8: Gemma-4-31B on Babylonian at $n_{\max} = 600$. Base-60 structure manifests primarily through its second harmonic at $T = 30$, indicating a sawtooth-like signal rather than a pure sinusoid.

basis omits, helix R^2 will be low even when the underlying structure is strong. The Babylonian result shows this concretely: same data, same model, $+0.13$ helix R^2 from adding one period (Table 7). The fix is to *always cross-check a basis-fit metric against a basis-free one*. If the FFT shows a peak at a period not in the basis, helix R^2 is undermeasuring the structure; add the period and refit.

M2. The window must wrap. To detect a period T via FFT, the input range must contain several wraps of T . One wrap is a single discontinuity, not a periodic signal. The 0–99 window is fine for $T \in \{2, 5, 10\}$ (50, 20, 10 wraps) and barely workable for $T = 100$ (one wrap, visible as a slow trend). It is *not* workable for $T = 60$: one wrap, in a phase-misaligned position relative to the window’s endpoints, produces noise rather than a trend. The rule is to set $n_{\max}$ to a large multiple of every candidate period.

These two corrections compose: on Babylonian at $n_{\max} = 100$ with the paper basis, both blind spots conspire in the same direction (“no base-60 here”). Fixing only one and not the other gives partial evidence; fixing both, and observing that the metrics move together and in the predicted direction, is what makes the inference robust.

7 Discussion and conclusion

Putting the threads together:

- **Cross-architecture:** the helix appears on Gemma-4-E4B and Gemma-4-31B—a new model family beyond the three (GPT-J, Pythia, Llama) the original paper tested—and re-measurement on Pythia and Llama under our wider protocol replicates the helix at very

different depths and with qualitatively different layer-profiles. “Read the middle layer” is a Pythia/GPT-J convention. Find-the-layer is a necessary step for every new model.

- **Cross-script, positional:** the geometric encoding transfers across glyph systems for any positional base-10 script the model has been trained on. Multilingual models reproduce the helix on Devanagari and Arabic-Indic; English-heavy models do not. The encoding is built per-script in pre-training; positional structure is necessary but not sufficient.
- **Cross-script, non-positional:** Greek and Roman both collapse the helix metric, but for different reasons. Roman splits into a number-line module and a letter-cycling module living in different layers; Greek never forms a quantitative representation at all (the only periodic structure in the FFT is a tokeniser artifact at $L = 0$). The two failure modes are distinguishable only by reading PC1 R^2 and its layer-of-peak alongside helix R^2 . The 2-D PCA scatter (Section 4.5) further reveals that Roman is encoded as discrete prefix-class clusters with crude ordering within each, and surfaces the most surprising finding of the script axis: on Gemma the first principal component of the residual stream captures up to 96.6% of variance but represents *tokenisation* (digit-count, leading-digit class), not magnitude. High PC1 R^2 is not by itself evidence of a magnitude axis; the helix in Gemma lives in lower-variance directions.
- **Cross-script, sexagesimal:** Babylonian cuneiform’s base 60 is represented on all four models—a stronger generalisation than the paper makes, lying entirely outside the representational space its measurement protocol can resolve. Recovering it required widening the window past one wrap and adding $T = 60$ to the trig basis.

We started this work expecting the helix to be a property of one model’s training. What appears truer now is that the helix is a property of how transformers encode anything they represent as a quantity, parameterised by which scripts the model has seen and which periods are natural to those scripts. Different models pick up different periods at different depths; the geometric strategy is the same. Whether the helix subspace is also *the representation that downstream computation reads* is a causal question this paper does not answer; the natural intervention sits at the answer position rather than the operand position, and we leave it to follow-up work.

7.1 Follow-ups

The natural next experiments are causal:

1. **Helix-subspace patching at the answer position.** The original paper’s strongest causal claim is that on Pythia/Latin, patching only the helix-basis subspace (~ 9 directions of several thousand) at the position whose hidden state produces the answer is sufficient to control the answer. Replicating that on Pythia and extending to Llama, Gemma-4-E4B and Gemma-4-31B—and to the non-Latin positional scripts where the geometric helix is present in this paper—is the most direct causal follow-up to the work reported here.
2. **Wider basis.** The helix basis used here is the paper’s $T \in \{2, 5, 10, 100\}$. Extending to the basis that recovers the Babylonian base-60 finding ($T \in \{2, 5, 10, 60, 100\}$), or to a *learned* subspace whose rank is set by cumulative explained-variance rather than by a prescribed period list, is the natural follow-up if a four-period helix-subspace intervention leaves a large unexplained fraction of the causal effect.
3. **Sexagesimal rotation.** On Gemma-4-31B’s Babylonian representation at $n_{\max} = 600$, rotating only the $(\cos 2\pi a/60, \sin 2\pi a/60)$ direction pair should shift the answer-token logit distribution by exactly 60 if the model uses the base-60 axis computationally. This is the cleanest test of representation-vs-use for a non-base-10 period.

4. **Cross-layer units-digit rescue.** For the E4B/Latin dissociation: project residual at L_5 onto $(\cos 2\pi a/10, \sin 2\pi a/10)$ and inject the projection at a later layer to see whether the units-digit information is discarded by downstream layers or remains accessible.

Secondary follow-ups suggested by the descriptive work but not addressed here: Greek alphabetic numerals plausibly form 9 clusters of 10, one per tens-letter, visible only via 2-D PCA scatter (the diagnostic is included but cluster analysis is left for future work); Roman’s encoding looks like a piecewise-linear staircase with discontinuities at glyph thresholds, which an additive regression $h(a) \approx \sum_{\text{symbol}} \text{count}(\text{symbol}) \cdot u_{\text{symbol}}$ would test directly; and the layer at which the Babylonian base-60 representation peaks, and how it differs from the Latin base-10 peak in the same model, is unaddressed in this paper and worth dedicated analysis.

Reproducibility

All code, configuration, and figures are in the public repository accompanying this paper. Each experiment lives in its own driver script with PEP 723 inline metadata, runnable via `uv run`; each has an accompanying `./run_*.sh` bash wrapper that loops over the four models and is idempotent (each combination is skipped if its expected output file already exists).

Representational sweep (Sections 3–5). `./run.sh` drives 40 combinations across 4 models, 8 scripts, 2 windows ($n_{\max} \in \{100, 600\}$), 2 bases ($T = \{2, 5, 10, 100\}$ and $T = \{2, 5, 10, 60, 100\}$):

```
# Default: Pythia-6.9B, Latin digits, paper basis, n=100
uv run main.py

# Cross-script
uv run main.py --model meta-llama/Llama-3.1-8B --script devanagari
uv run main.py --model google/gemma-4-31B --script babylonian --sweep

# Wider window
uv run main.py --script babylonian --n_max 600 --sweep

# Extended basis with T = 60
uv run main.py --script babylonian --n_max 600 \
    --periods 2,5,10,60,100 --sweep
```

Hardware: Apple M5 Max, 128 GB unified memory (MPS); falls back to CUDA, then CPU. Total wall-time for the representational sweep: ~ 8 hours.

References

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